centroids.

#9 Return the results (optionally you could save the clustering corresponding to the best metric value for each value of k, and return the list of clusterings as well).

12.5.1 Interpreting the Results

To check out if the elbow method works, we have stored the results producing the best metric values for each value of k tested (the ones that are plotted in figure 12.25) and we can take a look at some of them, in figure 12.26, to verify that our choice makes sense. And indeed, with k=5 we would have too few centroids, and two clusters would be assigned to the same centroid (left chart, bottom-left corner), while with k=7, a single “natural” cluster gets split by two centroids (right chart, middle).

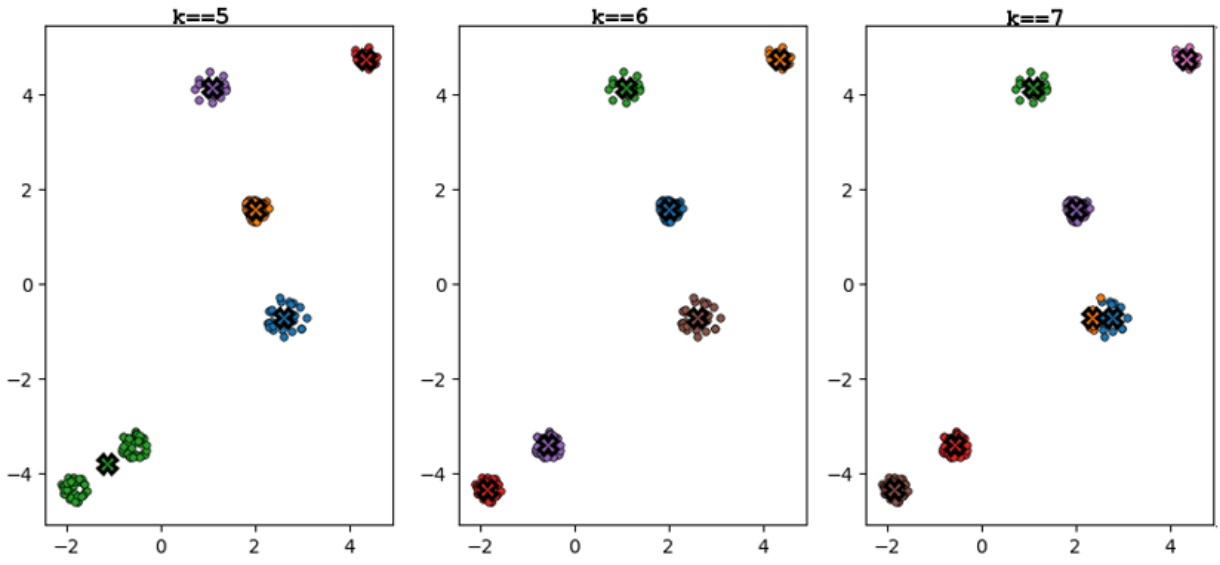


Figure 12.26 Best k-means clustering for our example dataset with k equal 5, 6 and 7. You can notice that 6